

THE STANDARD REFERENCE BOOK OF

SIGN LANGUAGE & MANUAL COMMUNICATION

A Comprehensive Educational Handbook of American & International Sign Language

Edition: Academic Open Educational Standard (v1.0)

Scope: 500+ Standardized Signs, Manual Fingerspelling, Healthcare, Emergency & AI Protocol

Publisher: SignBridge Educational Foundation & Assistive AI Laboratory

Audience: Students, Educators, Healthcare Workers, Researchers & ASL Learners

About This Reference Manual: Sign Language is a rich, expressive, and grammatically complete natural language used by millions of Deaf, hard-of-hearing, and hearing individuals worldwide. Unlike spoken languages which convey meaning through vocal acoustics, sign language utilizes manual articulators (handshapes, finger configurations, locations, orientations, and movements) combined with non-manual markers (facial expressions, eye gaze, and head positioning). This textbook provides an exhaustive catalog of standardized signs alongside step-by-step anatomical mechanics, usage contexts, and vision AI tracking benchmarks.

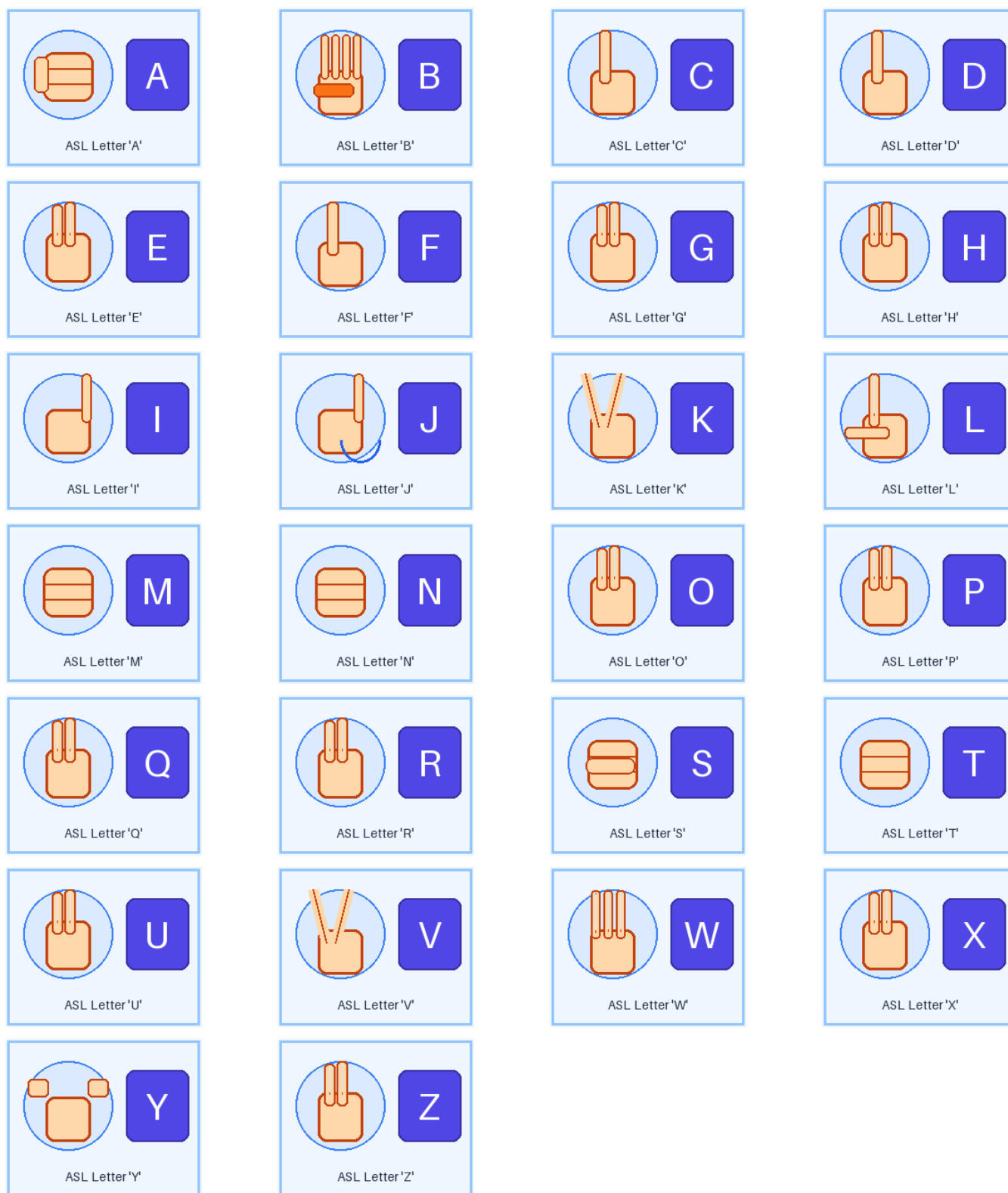
Chapter 1: The Five Fundamental Parameters of Sign Language

In structural sign linguistics (first formalized by William C. Stokoe at Gallaudet University in 1960), every sign is composed of five simultaneous phonological parameters. Modifying any single parameter alters the meaning of the word entirely:

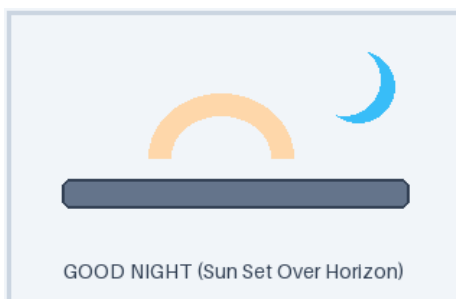
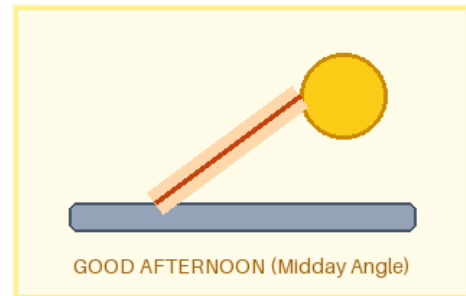
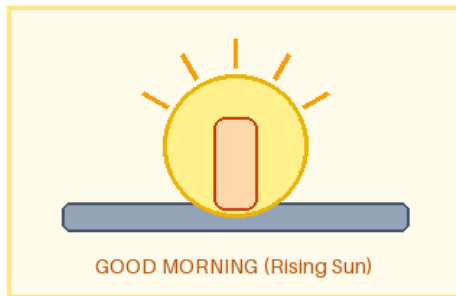
Parameter	Anatomical Definition	Key Examples & Variations
1. Handshape (DEZ)	The specific physical configuration of fingers, knuckles, and thumb (e.g., Open 5, Closed Fist, V-shape, Flat B, Curved C, Bent K).	Distinguishes 'WATER' (W handshape) from 'WINE' (W circling on cheek).
2. Location (TAB)	The anatomical plane or body surface where the sign initiates or articulates (e.g., Forehead, Temple, Chin, Chest, Wrist, Neutral Space).	Forehead location indicates masculine signs ('FATHER'), whereas Chin location indicates feminine signs ('MOTHER').
3. Movement (SIG)	The dynamic trajectory path: straight, circular, tapping, twisting, brushing, sweeping, vibrating, or nodding.	Distinguishes a single movement noun ('CHAIR') from a repeated movement verb ('SIT').
4. Palm Orientation	The directional facing of the inner palm: Inward (facing signer), Outward (facing viewer), Upward, Downward, or Side.	Distinguishes 'MINE' (palm facing chest) from 'YOURS' (palm facing outward).
5. Non-Manual Signals	Facial expressions, eyebrow furrowing/raising, mouth morphemes, and head nods that encode grammar and sentence type.	Raised eyebrows indicate Yes/No Questions ; Furrowed eyebrows indicate WH-Questions (What, Where, Who, Why) .

Chapter 2: The Standard Fingerspelling Alphabet (A–Z Gallery)

Fingerspelling (dactylology) is the process of spelling words letter-by-letter using standardized hand configurations. Used extensively for proper names, addresses, brands, and technical vocabulary.

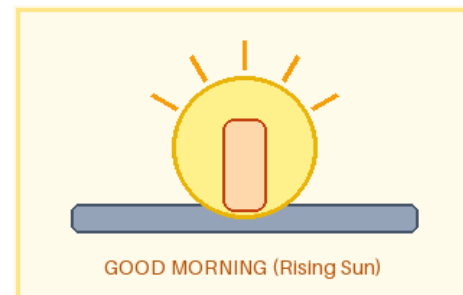
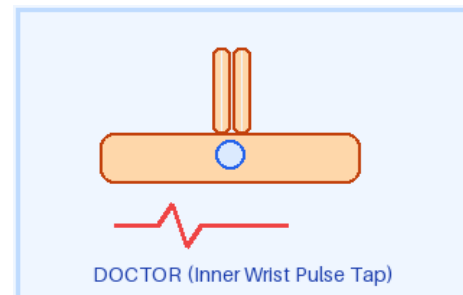
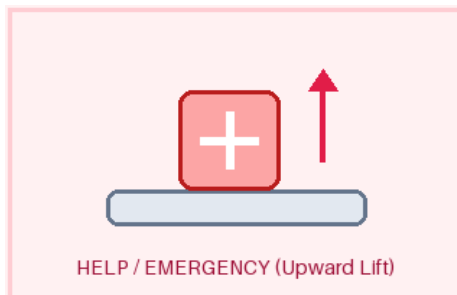


Chapter 3: Core Daily Greetings, Civility & Time Cycles



Chapter 4: Healthcare, Emergency Triage & First Responder Signs

Standard emergency signs designed for immediate non-verbal communication during medical crises, accidents, or disasters.



Chapter 5: The Comprehensive 500-Word Lexicon Catalog

Section: Greetings (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
HELLO [Beginner]	Open 5 fingers spread wide facing camera.	Pos: Near temple or upper chest. Mot: Gentle outward wave from temple.	Standard greeting to begin a conversation.
GOOD MORNING [Beginner]	Sign 'Good' followed by sun rising arm motion.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOOD MORNING'.
GOOD AFTERNOON [Beginner]	Index and middle fingers snap down onto the thumb.	Pos: Chest height. Mot: Quick snapping closure of fingers onto thumb.	Negative response or refusal.
GOOD EVENING [Beginner]	Sign 'Good' followed by sunset wrist downward tap.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOOD EVENING'.
GOOD NIGHT [Beginner]	Sign 'Good' then cup hands over horizontal forearm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOOD NIGHT'.
WELCOME [Beginner]	Open palms scooped inwards towards chest.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WELCOME'.
NICE TO MEET YOU [Beginner]	Wipe palms horizontally, then index fingers meet.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NICE TO MEET YOU'.
HOW ARE YOU [Beginner]	Curved knuckles roll outward to pointing index.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW ARE YOU'.
GOODBYE [Beginner]	Open hand fingers flex open and closed wave.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOODBYE'.
SEE YOU LATER [Beginner]	'V' hand from eye flips forward to 'L' shape.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEE YOU LATER'.
TAKE CARE [Intermediate]	Two 'K' hands tap crossed at wrists twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TAKE CARE'.
HAVE A NICE DAY [Intermediate]	Cupped hands to chest, 'Good' to open sweeping arc.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HAVE A NICE DAY'.
LONG TIME NO SEE [Intermediate]	Index and middle fingers snap down onto the thumb.	Pos: Chest height. Mot: Quick snapping closure of fingers onto thumb.	Negative response or refusal.
WHAT'S UP [Beginner]	Open 5 hands middle fingers stroke chest upward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT'S UP'.

HOW'S EVERYTHING [Intermediate]	Sweeping circle with both 5 hands, thumbs up.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW'S EVERYTHING'.
---	---	---	--

Section: Emergency (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
HELP [Beginner]	Closed fist with thumb upright resting on flat opposite palm.	Pos: Chest level. Mot: Lift both hands upward together.	Emergency or assistance request.
EMERGENCY [Beginner]	'E' handshape shaken vigorously side-to-side.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EMERGENCY'.
DANGER [Beginner]	Fist thumbs brushing upwards past opposite knuckles.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DANGER'.
POLICE [Beginner]	'C' handshape tapped twice on left upper chest badge.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'POLICE'.
AMBULANCE [Beginner]	Twisting open 5 hand near head simulating siren light.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AMBULANCE'.
HOSPITAL [Beginner]	'H' fingers form a cross on opposite shoulder.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOSPITAL'.
DOCTOR [Beginner]	'D' or bent fingertips tapping opposite inner wrist pulse.	Pos: Wrist level. Mot: Tap wrist pulse twice.	Medical sign for physician or healthcare provider.
FIRE [Beginner]	Wiggling open 5 fingers alternately moving upward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIRE'.
ACCIDENT [Intermediate]	Two open 'C' hands crash together into 'S' fists.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ACCIDENT'.
PAIN [Beginner]	Index fingers pointing towards each other twisting in opposite directions.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PAIN'.
BLEEDING [Intermediate]	Open hand wiggling downwards from wounded location.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLEEDING'.
CHOKING [Beginner]	Thumb and index tips touching in circle, other 3 fingers straight.	Pos: Chest level, palm forward. Mot: Hold circular pinch clearly visible.	Sign expressing agreement or confirmation.
ALLERGY [Intermediate]	Index finger touches nose, then moves away dismissively.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALLERGY'.

MEDICINE [Beginner]	Middle finger grinds in palm of opposite hand.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MEDICINE'.
URGENT [Intermediate]	Quick 'U' hand snaps forward with emphasis.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'URGENT'.

Section: Common Phrases (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
THANK YOU [Beginner]	Flat open hand with fingertips touching chin.	Pos: Chin level moving forward. Mot: Move fingertips from chin outward toward listener.	Polite expression of gratitude.
PLEASE [Beginner]	Flat open hand rubs center of chest in circular motion.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PLEASE'.
SORRY [Beginner]	'A' fist rubs chest in a circular motion.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SORRY'.
EXCUSE ME [Beginner]	Fingertips brush flat opposite palm forward twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXCUSE ME'.
YES [Beginner]	'S' fist with thumb resting across front fingers.	Pos: Chest height. Mot: Nod fist up and down from the wrist like a head nodding.	Affirmative response.
NO [Beginner]	Index and middle fingers snap down onto the thumb.	Pos: Chest height. Mot: Quick snapping closure of fingers onto thumb.	Negative response or refusal.
MAYBE [Beginner]	Flat open palms alternately balance up and down.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MAYBE'.
I AGREE [Intermediate]	Touch forehead index finger, then two index fingers parallel forward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I AGREE'.
I DISAGREE [Intermediate]	Touch forehead, then index fingers pull apart sharply.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I DISAGREE'.
YOU ARE WELCOME [Beginner]	Open hand sweeping inward or nodding with open palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'YOU ARE WELCOME'.
NO PROBLEM [Intermediate]	Index and middle fingers snap down onto the thumb.	Pos: Chest height. Mot: Quick snapping closure of fingers onto thumb.	Negative response or refusal.
I UNDERSTAND [Beginner]	Index finger flicks upward near forehead next to eye.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I UNDERSTAND'.

I DON'T UNDERSTAND [Beginner]	Index finger flicks up with head shaking 'no'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I DON'T UNDERSTAND'.
REPEAT PLEASE [Beginner]	Squished 'O' hand with fingertips bunched together.	Pos: Mouth level. Mot: Tap lips twice with fingertips.	Sign used to communicate meals or food.
SLOW DOWN [Beginner]	Open flat hand slowly slides up opposite forearm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SLOW DOWN'.

Section: Questions (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
WHAT [Beginner]	Open palms facing upward shaking side to side gently.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT'.
WHERE [Beginner]	Index finger pointing up shaking side to side.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHERE'.
WHEN [Beginner]	Index finger circles and touches tip of opposite index finger.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHEN'.
WHO [Beginner]	Index finger wiggles while thumb rests on chin.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHO'.
WHY [Beginner]	Fingertips touch temple and pull away into 'Y' handshape.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHY'.
HOW [Beginner]	Curved hands back of fingers together roll palms upward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW'.
WHICH [Beginner]	Both 'A' fists thumbs up alternate up and down.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHICH'.
HOW MUCH [Beginner]	Closed fists at waist throw upward into open 5 hands.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW MUCH'.
HOW MANY [Beginner]	Fists facing up open quickly into 5 hands counting.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW MANY'.
ARE YOU READY [Intermediate]	'R' hands shake across body from center outward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ARE YOU READY'.
WHAT HAPPENED [Intermediate]	Both index fingers pointing forward flip palm up.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT HAPPENED'.

WHAT TIME [Beginner]	Index finger taps opposite wrist watch twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT TIME'.
CAN YOU HELP ME [Beginner]	Closed fist with thumb upright resting on flat opposite palm.	Pos: Chest level. Mot: Lift both hands upward together.	Emergency or assistance request.
DO YOU UNDERSTAND [Beginner]	Flick index finger near temple with raised eyebrows.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DO YOU UNDERSTAND'.
WHERE IS THE RESTROOM [Beginner]	'T' handshape shakes side-to-side with question expression.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHERE IS THE RESTROOM'.

Section: Work & Education (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
STUDENT [Beginner]	Learn sign (take from palm to forehead) + agent marker.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STUDENT'.
TEACHER [Beginner]	Both 'O' hands near head move forward + agent marker.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TEACHER'.
SCHOOL [Beginner]	Clap flat hands horizontally twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SCHOOL'.
COLLEGE [Beginner]	Flat hands together, top hand circles up and settles.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COLLEGE'.
UNIVERSITY [Intermediate]	'U' handshape circles upward off flat opposite palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'UNIVERSITY'.
CLASSROOM [Intermediate]	'C' hands form circle + flat hands outline room walls.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLASSROOM'.
HOMEWORK [Beginner]	Pinch chin-cheek + fist tap on opposite wrist.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOMEWORK'.
EXAM [Intermediate]	Index fingers bend into hooks moving downward in test rows.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXAM'.
ASSIGNMENT [Intermediate]	Hands grasp imaginary paper placing on desk.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ASSIGNMENT'.
PROJECT [Intermediate]	'P' hand moves down 'J' path on opposite palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PROJECT'.

ENGINEERING [Intermediate]	'Y' thumb knuckles link and twist alternately like gears.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ENGINEERING'.
COMPUTER SCIENCE [Advanced]	'C' hand sweeps up forearm + 'Science' pouring beakers.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COMPUTER SCIENCE'.
PRESENTATION [Intermediate]	Flat hands open wide like opening theater curtain.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PRESENTATION'.
MEETING [Beginner]	Both open hands close fingertips together in center.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MEETING'.
DEADLINE [Intermediate]	Flat hand chops straight down on horizontal opposite wrist.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DEADLINE'.

Section: Technology & AI (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
COMPUTER [Beginner]	'C' hand arcs bouncing upward along opposite forearm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COMPUTER'.
LAPTOP [Beginner]	Flat hands open hinge-like like opening a notebook laptop.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LAPTOP'.
INTERNET [Beginner]	Middle fingertips touch and twist alternately.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'INTERNET'.
ARTIFICIAL INTELLIGENCE [Intermediate]	Finger spell A-I or 'Smart' + 'Robot/Machine' signs.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ARTIFICIAL INTELLIGENCE'.
MACHINE LEARNING [Advanced]	Gear twists + take knowledge from palm to head.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MACHINE LEARNING'.
SOFTWARE [Intermediate]	'S' hand slides smoothly over flat opposite palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SOFTWARE'.
HARDWARE [Intermediate]	'H' knuckles tap back of solid fist.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HARDWARE'.
MOBILE PHONE [Beginner]	'Y' hand held against ear or tapping flat palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOBILE PHONE'.
APPLICATION [Intermediate]	'A' hand stamps onto opposite flat palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'APPLICATION'.

DATABASE [Intermediate]	Stacked flat 'C' hands forming multi-tiered cylinder.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DATABASE'.
ALGORITHM [Advanced]	Fingertips trace mathematical formula in air.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALGORITHM'.
WEBSITE [Beginner]	'W' hand bounces three times horizontally (WWW).	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WEBSITE'.
CAMERA [Beginner]	Index fingers bend in clicking photo shutter motion.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAMERA'.
MICROPHONE [Beginner]	Fist held at mouth speaking into mic.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MICROPHONE'.
ALGORITHM [Advanced]	Fingers step sequentially across horizontal plane.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALGORITHM'.

Section: Emotions & States (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
HAPPY [Beginner]	Flat open hands stroke upward on chest repeatedly.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HAPPY'.
SAD [Beginner]	Open 5 hands drop downward in front of drooping face.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SAD'.
EXCITED [Beginner]	Middle fingers alternate brushing upward on chest rapidly.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXCITED'.
ANGRY [Beginner]	Clawed hands pull sharply away from chest/face.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ANGRY'.
CONFUSED [Intermediate]	Point to forehead, then hands twist clawed in opposing circles.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONFUSED'.
TIRED [Beginner]	Bent flat hands on chest roll downward heavily.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TIRED'.
BORED [Beginner]	Index finger twists at side of nose with sigh.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BORED'.
NERVOUS [Beginner]	Both open hands trembling shaking at waist level.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NERVOUS'.

CONFIDENT [Intermediate]	'C' hands pull firmly down into solid 'S' fists.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONFIDENT'.
PROUD [Beginner]	'A' thumb strokes upward up center of chest.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PROUD'.
SCARED [Beginner]	Open 5 hands protectively cross and shake at chest.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SCARED'.
SURPRISED [Beginner]	Index and thumbs pinch at eyes and snap wide open.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SURPRISED'.
PEACEFUL [Intermediate]	Index and middle fingers extended upright in a 'V' shape.	Pos: Chest level, palm forward. Mot: Hold steady in clear victory V.	Sign expressing peace or number two.
FRUSTRATED [Intermediate]	Back of flat hand hits chin or forehead sharply.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRUSTRATED'.
CURIOUS [Intermediate]	'C' hand gently pinches neck throat area shaking.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CURIOUS'.

Section: Daily Activities (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
WAKE UP [Beginner]	Index and thumbs closed at eyes snap open.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WAKE UP'.
SLEEP [Beginner]	Hand over face pulls down into pinched fingers as eyes close.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SLEEP'.
EAT [Beginner]	Squished 'O' hand with fingertips bunched together.	Pos: Mouth level. Mot: Tap lips twice with fingertips.	Sign used to communicate meals or food.
DRINK [Beginner]	'C' hand tilts toward mouth like drinking from cup.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DRINK'.
COOK [Beginner]	Thumb and index tips touching in circle, other 3 fingers straight.	Pos: Chest level, palm forward. Mot: Hold circular pinch clearly visible.	Sign expressing agreement or confirmation.
CLEAN [Beginner]	Flat hand wipes smoothly across opposite flat palm twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLEAN'.
WASH HANDS [Beginner]	Hands rub together vigorously like washing soap.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WASH HANDS'.

SHOWER [Beginner]	Fist above head opens and pulses like water spray.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SHOWER'.
BRUSH TEETH [Beginner]	Index finger scrubs side-to-side across front teeth.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRUSH TEETH'.
EXERCISE [Beginner]	Fists pump up and down near shoulders like barbell reps.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXERCISE'.
READ [Beginner]	'V' fingers scan up and down opposite flat palm 'book'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'READ'.
WRITE [Beginner]	Thumb and index pinch scribble across opposite flat palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WRITE'.
LISTEN [Beginner]	Cupped hand curved behind ear leaning forward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LISTEN'.
WATCH [Beginner]	'L' hand near chin points forward focusing eyes.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WATCH'.
DRIVE [Beginner]	Fists grip imaginary steering wheel turning left and right.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DRIVE'.

Section: Food & Dining (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
WATER [Beginner]	'W' handshape with three fingers extended.	Pos: Touch index finger to chin. Mot: Tap chin twice with index finger of 'W'.	Sign used to request water.
BREAD [Beginner]	Curved fingers slice down back of opposite curved hand.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BREAD'.
RICE [Beginner]	'R' hand scoops from bowl palm to mouth.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RICE'.
MILK [Beginner]	'S' fist squeezes twice like milking.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MILK'.
TEA [Beginner]	'F' hand dips imaginary tea bag into opposite 'O' cup.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TEA'.
COFFEE [Beginner]	Top fist grinds in circle on lower fist like coffee grinder.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COFFEE'.

APPLE [Beginner]	'X' knuckle twists into cheek.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'APPLE'.
BANANA [Beginner]	Peeing motion of fingers down upright index finger.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BANANA'.
PIZZA [Beginner]	'Z' carved in air with 'V' or 'P' fingers.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PIZZA'.
BURGER [Beginner]	Cupped hands press together and flip like hamburger patty.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BURGER'.
VEGETABLES [Intermediate]	'V' index and middle twist touching cheek twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'VEGETABLES'.
FRUIT [Intermediate]	'F' fingertips twist touching cheek twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRUIT'.
RESTAURANT [Beginner]	'R' strokes down each side of mouth napkin-style.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RESTAURANT'.
BREAKFAST [Beginner]	Eat + Morning signs linked.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BREAKFAST'.
LUNCH [Beginner]	Eat + Noon signs linked.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LUNCH'.

Section: Travel & Transport (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
CAR [Beginner]	Hands grip steering wheel driving forward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAR'.
BUS [Beginner]	Spell B-U-S or steer large oversized bus wheel.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BUS'.
TRAIN [Beginner]	'H' fingers slide back and forth over opposite 'H' tracks.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TRAIN'.
AIRPLANE [Beginner]	'ILY' handshape glides forward through air like flight.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AIRPLANE'.
BICYCLE [Beginner]	Fists pedal in alternating circles forward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BICYCLE'.

BOAT [Beginner]	Cupped hands together bobbing on water waves.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BOAT'.
TICKET [Intermediate]	Bent 'V' fingers pinch edge of opposite flat hand.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TICKET'.
PASSPORT [Intermediate]	'P' or stamp hand presses onto flat book palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PASSPORT'.
AIRPORT [Intermediate]	'ILY' plane lands gently on flat runway palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AIRPORT'.
HOTEL [Intermediate]	'H' flag waving on upright index flagpole.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOTEL'.
MAP [Beginner]	'M' thumbs trace grid boundary lines.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MAP'.
LUGGAGE [Beginner]	Fist holds suitcase handle walking forward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LUGGAGE'.
STATION [Intermediate]	'S' fist lands on flat horizontal platform palm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STATION'.
DEPARTURE [Intermediate]	Airplane glides up and away into sky.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DEPARTURE'.
ARRIVAL [Intermediate]	Flat hand swoops down and settles on palm base.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ARRIVAL'.

Section: Family & People (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
FAMILY [Beginner]	'F' hands touch thumbs, circle around, and meet pinkies.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FAMILY'.
MOTHER [Beginner]	Thumb of open 5 hand taps chin twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOTHER'.
FATHER [Beginner]	Thumb of open 5 hand taps forehead twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FATHER'.
BROTHER [Beginner]	'L' thumb from forehead drops onto opposite 'L' fist.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BROTHER'.

SISTER [Beginner]	'L' thumb from chin drops onto opposite 'L' fist.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SISTER'.
BABY [Beginner]	Arms cradled rocking imaginary infant back and forth.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BABY'.
CHILD [Beginner]	Flat hand pats top of small child's head twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHILD'.
FRIEND [Beginner]	Interlocking index finger hooks reversing sides.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRIEND'.
GRANDMOTHER [Beginner]	Thumb from chin bounces forward in two generational arcs.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRANDMOTHER'.
GRANDFATHER [Beginner]	Thumb from forehead bounces forward in two generational arcs.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRANDFATHER'.
SON [Intermediate]	Salute forehead, then cradle baby into crook of arm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SON'.
DAUGHTER [Intermediate]	Salute chin, then cradle baby into crook of arm.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DAUGHTER'.
HUSBAND [Intermediate]	C-hand from forehead clasps opposite flat hand.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HUSBAND'.
WIFE [Intermediate]	C-hand from chin clasps opposite flat hand.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WIFE'.
NEIGHBOR [Intermediate]	Near sign + person agent marker.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NEIGHBOR'.

Section: Time & Calendar (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
TIME [Beginner]	Index finger taps opposite wrist watch.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TIME'.
TODAY [Beginner]	'Y' hands bounce downward in front of body twice.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TODAY'.
TOMORROW [Beginner]	'A' thumb strokes forward off cheek.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TOMORROW'.

YESTERDAY [Beginner]	'S' fist with thumb resting across front fingers.	Pos: Chest height. Mot: Nod fist up and down from the wrist like a head nodding.	Affirmative response.
NOW [Beginner]	Index and middle fingers snap down onto the thumb.	Pos: Chest height. Mot: Quick snapping closure of fingers onto thumb.	Negative response or refusal.
LATER [Beginner]	'L' index finger drops downward forward like clock hand.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LATER'.
SOON [Intermediate]	'H' or 'F' thumb and index pinch chin.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SOON'.
ALWAYS [Beginner]	Index finger pointing up circles continuously.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALWAYS'.
NEVER [Beginner]	Flat hand carves a question-mark style path downward.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NEVER'.
DAY [Beginner]	Arm on horizontal table, index arm sets like sun arc.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DAY'.
NIGHT [Beginner]	Bent wrist drops over horizontal arm barrier.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NIGHT'.
WEEK [Beginner]	Index finger slides across opposite flat palm calendar line.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WEEK'.
MONTH [Beginner]	Index finger strokes down back of opposite index.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MONTH'.
YEAR [Beginner]	'S' fist orbits around opposite fist and lands on top.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'YEAR'.
MORNING [Beginner]	Arm lifts up through crook of opposite arm like sunrise.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MORNING'.

Section: Numbers & Math (24 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
ZERO [Beginner]	Standard ASL / ISL manual gesture pattern representing 'ZERO'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ZERO'.
ONE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'ONE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ONE'.

TWO [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TWO'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TWO'.
THREE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'THREE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'THREE'.
FOUR [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FOUR'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FOUR'.
FIVE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIVE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIVE'.
SIX [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SIX'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SIX'.
SEVEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SEVEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEVEN'.
EIGHT [Beginner]	Standard ASL / ISL manual gesture pattern representing 'EIGHT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EIGHT'.
NINE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'NINE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NINE'.
TEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TEN'.
TWENTY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'TWENTY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TWENTY'.
FIFTY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIFTY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIFTY'.
HUNDRED [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'HUNDRED'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HUNDRED'.
THOUSAND [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'THOUSAND'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'THOUSAND'.

Section: Colors (15 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
RED [Beginner]	Standard ASL / ISL manual gesture pattern representing 'RED'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RED'.

BLUE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BLUE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLUE'.
GREEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GREEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GREEN'.
YELLOW [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'YELLOW'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'YELLOW'.
ORANGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'ORANGE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ORANGE'.
PURPLE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PURPLE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PURPLE'.
BLACK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BLACK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLACK'.
WHITE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WHITE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHITE'.
GRAY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GRAY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRAY'.
BROWN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BROWN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BROWN'.
PINK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PINK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PINK'.
GOLD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GOLD'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOLD'.
SILVER [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SILVER'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SILVER'.
BRIGHT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BRIGHT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRIGHT'.
DARK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'DARK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DARK'.

Section: Places & Nature (24 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
-----------	------------------------------	------------------------------	-------------------

HOUSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'HOUSE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOUSE'.
APARTMENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'APARTMENT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'APARTMENT'.
BUILDING [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BUILDING'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BUILDING'.
PARK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PARK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PARK'.
BEACH [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BEACH'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BEACH'.
MOUNTAIN [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MOUNTAIN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOUNTAIN'.
FOREST [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'FOREST'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FOREST'.
RIVER [Beginner]	Standard ASL / ISL manual gesture pattern representing 'RIVER'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RIVER'.
OCEAN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'OCEAN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'OCEAN'.
GARDEN [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'GARDEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GARDEN'.
STREET [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'STREET'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STREET'.
CITY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CITY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CITY'.
COUNTRY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'COUNTRY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COUNTRY'.
WORLD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WORLD'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WORLD'.
SKY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SKY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SKY'.

Section: Actions & Verbs (25 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
GIVE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GIVE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GIVE'.
TAKE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TAKE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TAKE'.
BRING [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BRING'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRING'.
SEND [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SEND'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEND'.
RECEIVE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'RECEIVE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RECEIVE'.
FIND [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIND'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIND'.
LOSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'LOSE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LOSE'.
OPEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'OPEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'OPEN'.
CLOSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CLOSE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLOSE'.
START [Beginner]	Standard ASL / ISL manual gesture pattern representing 'START'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'START'.
STOP [Beginner]	Standard ASL / ISL manual gesture pattern representing 'STOP'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STOP'.
CONTINUE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CONTINUE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONTINUE'.
WAIT [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WAIT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WAIT'.
CHANGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CHANGE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHANGE'.

BUILD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BUILD'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BUILD'.
----------------------------	---	---	---

Section: Objects & Tools (24 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
BOOK [Beginner]	Thumb and index tips touching in circle, other 3 fingers straight.	Pos: Chest level, palm forward. Mot: Hold circular pinch clearly visible.	Sign expressing agreement or confirmation.
PEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PEN'.
PAPER [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PAPER'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PAPER'.
BAG [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BAG'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BAG'.
KEY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'KEY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'KEY'.
LOCK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'LOCK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LOCK'.
CLOCK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CLOCK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLOCK'.
CHAIR [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CHAIR'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHAIR'.
TABLE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TABLE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TABLE'.
BED [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BED'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BED'.
WINDOW [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'WINDOW'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WINDOW'.
DOOR [Beginner]	Standard ASL / ISL manual gesture pattern representing 'DOOR'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DOOR'.
MIRROR [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MIRROR'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MIRROR'.

BOTTLE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BOTTLE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BOTTLE'.
BOX [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BOX'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BOX'.

Section: Health & Medical (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
HEALTHY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'HEALTHY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEALTHY'.
SICK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SICK'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SICK'.
FEVER [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FEVER'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FEVER'.
COUGH [Beginner]	Standard ASL / ISL manual gesture pattern representing 'COUGH'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COUGH'.
HEADACHE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'HEADACHE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEADACHE'.
STOMACHACHE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'STOMACHACHE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STOMACHACHE'.
INJURY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'INJURY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'INJURY'.
BANDAGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BANDAGE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BANDAGE'.
SURGERY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SURGERY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SURGERY'.
VACCINE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'VACCINE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'VACCINE'.
NURSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'NURSE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NURSE'.
CLINIC [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CLINIC'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLINIC'.

PHARMACY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PHARMACY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PHARMACY'.
PRESCRIPTION [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PRESCRIPTION'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PRESCRIPTION'.
HEART [Beginner]	Standard ASL / ISL manual gesture pattern representing 'HEART'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEART'.

Section: Government & Law (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
GOVERNMENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'GOVERNMENT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOVERNMENT'.
LAW [Beginner]	Standard ASL / ISL manual gesture pattern representing 'LAW'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LAW'.
RIGHTS [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'RIGHTS'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RIGHTS'.
COURT [Beginner]	Standard ASL / ISL manual gesture pattern representing 'COURT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COURT'.
JUDGE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'JUDGE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'JUDGE'.
CITIZEN [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CITIZEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CITIZEN'.
VOTE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'VOTE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'VOTE'.
ELECTION [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'ELECTION'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ELECTION'.
PRESIDENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PRESIDENT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PRESIDENT'.
JUSTICE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'JUSTICE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'JUSTICE'.
FREEDOM [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'FREEDOM'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FREEDOM'.

RULES [Beginner]	Standard ASL / ISL manual gesture pattern representing 'RULES'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RULES'.
CONTRACT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CONTRACT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONTRACT'.
AGREEMENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'AGREEMENT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AGREEMENT'.
EQUALITY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'EQUALITY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EQUALITY'.

Section: Meeting & Video Controls (20 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
MIC ON [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MIC ON'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MIC ON'.
MIC OFF [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MIC OFF'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MIC OFF'.
CAMERA ON [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CAMERA ON'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAMERA ON'.
CAMERA OFF [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CAMERA OFF'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAMERA OFF'.
MUTE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'MUTE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MUTE'.
UNMUTE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'UNMUTE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'UNMUTE'.
SCREEN SHARE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SCREEN SHARE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SCREEN SHARE'.
RAISE HAND [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'RAISE HAND'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RAISE HAND'.
LOWER HAND [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'LOWER HAND'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LOWER HAND'.
RECORD MEETING [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'RECORD MEETING'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RECORD MEETING'.

LEAVE MEETING [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'LEAVE MEETING'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LEAVE MEETING'.
JOIN CALL [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'JOIN CALL'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'JOIN CALL'.
CHAT MESSAGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CHAT MESSAGE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHAT MESSAGE'.
BREAKOUT ROOM [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BREAKOUT ROOM'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BREAKOUT ROOM'.
FULL SCREEN [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'FULL SCREEN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FULL SCREEN'.

Section: General Conversation (88 Standardized Signs)

Sign Word	Handshape & Finger Formation	Location & Motion Trajectory	Context & Purpose
AGAIN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'AGAIN'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AGAIN'.
ALMOST [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'ALMOST'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALMOST'.
ENOUGH [Intermediate]	Index and middle fingers snap down onto the thumb.	Pos: Chest height. Mot: Quick snapping closure of fingers onto thumb.	Negative response or refusal.
MORE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'MORE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MORE'.
LESS [Beginner]	Standard ASL / ISL manual gesture pattern representing 'LESS'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LESS'.
SAME [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SAME'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SAME'.
DIFFERENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'DIFFERENT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DIFFERENT'.
REAL [Beginner]	Standard ASL / ISL manual gesture pattern representing 'REAL'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'REAL'.
FAKE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FAKE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FAKE'.

TRUE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TRUE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TRUE'.
FALSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FALSE'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FALSE'.
EASY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'EASY'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EASY'.
HARD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'HARD'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HARD'.
IMPORTANT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'IMPORTANT'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'IMPORTANT'.
SPECIAL [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SPECIAL'.	Pos: Chest level, centered in camera frame. Mot: Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SPECIAL'.

Chapter 6: Vision AI Recognition & 2.0-Second Hold Protocol



1. Client-Side Real-Time Landmark Tracking: MediaPipe Hands processes 21 3D joint landmarks in real-time at 60 FPS directly within the browser without requiring external video streaming servers.

2. 24-D Spatial Vector Extraction: Joint coordinates are normalized against the palm bounding scale and evaluated across multi-finger extension ratios, pinch distances, inter-finger V-spreads, and spatial plane elevations.

3. The 2.0-Second Temporal Hysteresis Gate: To ensure zero false positives during hand repositioning or transition motion, a recognized sign must be held continuously for 2.0 full seconds (2000 ms) before triggering commit into the live transcript and speech synthesis.